JVerb

The three-body reverb. User Manual



VERSION	FORMAT	PLATFORMS	EDITION
0.7.0	CLAP · Stereo	macOS · Windows · Linux	October 2026

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About this manual

Chapters 2 and 3 get you installed and making sound in a few minutes. Chapter 4 tours the panel, and Chapter 5 explains how the sound is made, which is the quickest way to understand what every control does. Chapters 6 to 9 cover every control in the order they sit on the panel. The appendix lists every parameter with its exact range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **DAMP HIGH**. Tags beside a parameter mean:

MOD — The host can modulate it with CLAP parameter modulation, on top of automation.

TARGET — A body can be routed to it as a modulation target.

CHAPTER 01

Welcome to JVerb

A reverb whose tail is moved by a live gravity simulation. Two to six bodies orbit one another, and each one is a voice inside the sound.

Most reverbs that move do it with an LFO: a sine wave, a fixed rate, the same swirl every bar. JVerb moves its tail with something that never quite repeats. At its heart is a lush sixteen-line reverb tank. Those lines are shared out among a small system of bodies that pull on each other under gravity, and each body's share of the tail leaves through its own **voice**: its own band of the spectrum, its own place in the stereo field, its own tone and its own phaser. As the bodies swing, slingshot and drift, the tail swirls around the speakers with them.

Three bodies start on the famous figure-eight orbit and slowly fall into chaos. Add more, up to six, and the motion gets busier. Drag them, throw them, make one heavy and the others light, or let the music kick them around: transients in your input nudge the bodies, so what you play changes the orbit, and the orbit changes the sound.

Around the tank sits an **outer loop** with feedback, shimmer, smear and tilt; a five-flavour **drive** you can place before, inside or after the reverb; and a four-second **echo** that can feed the reverb, be fed by it, or ride inside its loop.

At a glance

- **Sixteen-line tank** with two allpasses per line and a Hadamard mix. Decays from 0.2 to 60 seconds.
- **Two to six orbiting bodies**, each its own voice: band, pan, tone and swirl, driven by its position and angle.
- **Bodies as modulators.** Route any body's speed and close passes to Decay, Damping, Feedback, Width and more.
- **Reactive orbits.** Input transients kick the bodies by band: lows, mids, highs.
- **Outer loop** with Feedback to 150%, ±24-semitone shimmer, smear, tilt and Freeze.
- **Drive**: Tape, Tube, Fuzz, Fold or Crush, 2× oversampled, in the PRE, LOOP or POST slot.
- **Echo** up to 4 s, free or tempo-synced, with ping-pong, in the INTO, LOOP or AFTER slot.
- **An extreme zone** past 100% on four knobs, for warble, ringing and inside-out stereo.
- **32 factory presets**, from tight booths to minute-long washes.

> **LOVELY BELOW 100%, WEIRD ABOVE IT**
> Motion, Orbit Pan, Orbit Tone and Orbit Swirl run to 200%, with a tick on the knob at 100%. Up to the tick they are tuned to stay musical: nothing bends pitch audibly and nothing rings. Past the tick is the extreme zone. See *Chapter 12*.

CHAPTER 02

Installation

JVerb is a CLAP plug-in. It ships in the JClap bundle alongside the other JClap plug-ins.

Requirements

Plug-in format	CLAP 1.2 (audio effect, stereo). Your host must support CLAP, for example Bitwig Studio, REAPER, FL Studio or Studio One.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11.

Installing

01 **Download**

Grab the zip for your operating system from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. It contains every JClap plug-in.

02 **Copy the plug-in**

Copy `JVerb.clap` (a bundle on macOS) into your CLAP folder:

macOS	`~/Library/Audio/Plug-Ins/CLAP`
Windows	`C:\Program Files\Common Files\CLAP`
Linux	`~/.clap`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. JVerb appears as an audio effect in the reverb category.

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. The simplest install is from Terminal, since files fetched with `curl` are not flagged:

```
curl -sL https://github.com/jeffmhopkins/Jclap/releases/latest/download/jclap-macOS.zip -o /tmp/jclap.zip
&& mkdir -p ~/Library/Audio/Plug-Ins/CLAP && ditto -x -k /tmp/jclap.zip ~/Library/Audio/Plug-Ins/CLAP
```

If you already downloaded with a browser, clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap
```

Insert or send?

JVerb works either way. On an **insert**, use **MIX** in the Output section to set the balance of dry and wet (the default is 35%). On a **send/return** bus, set **MIX** to 100% so the return carries only the reverb, and set the amount with the send level instead. If you use **DUCK** on a send, remember that it listens to whatever arrives at JVerb's input, which is the send signal.

JVerb is a stereo effect. A mono input is used on both sides, and the reverb spreads it into stereo.

CHAPTER 03

Quick Start

Five minutes from an empty insert slot to a moving, living tail.

01 **Load JVerb and play something**

Put JVerb on a track with something sustained and something percussive in it: a pad and a drum loop are ideal. The orbits start running as soon as the panel opens. The bodies trail coloured paths across the stage, and each drum hit blows a puff of glowing gas out of them.

02 **Pick a preset**

Click the preset name in the header to open the menu: presets are grouped from *Short* rooms through *Medium* and *Long* spaces to *Washes* that ring for most of a minute. Or click the arrows, or scroll over the bar, to step through them. Try *Vocal Hall*, then *Shimmer Halo*, then *Nebula*.

03 **Set the size and length**

On the **SPACE** page, **DECAY** sets how long the tail lasts and **SYSTEM SIZE** how big the space is. Bigger systems have longer lines and a slower, grainier build. **DAMP HIGH** darkens the tail as it rings.

04 **Make it move**

Open the **ORBIT** page. **PAN** moves each body's voice from a wide spread to where the body actually is; **TONE** darkens voices as their bodies swing out; **SWIRL** adds a phaser to each voice, swept by its orbit. Now drag a body somewhere new, or scroll on it to make it heavier, and listen to the tail rearrange itself.

05 **Let a body drive something**

Each body has a card along the bottom. Set Kepler's target menu to **Feedback** and raise its **AMOUNT**: each time Kepler slingshots past another body, the tail surges. The Feedback knob's ring shows the movement live.

06 **Add colour on the right**

The right card has three pages. **LOOP** adds feedback and shimmer around the whole tank, **DRIVE** saturates, and **ECHO** adds rhythmic repeats. Try **ECHO ON** with the slot at **INTO**: every repeat blooms into its own reverb swell.

07 **Save it**

Click **SAVE** in the preset bar, type a name and press Enter. Your preset appears under *User* in the preset menu. Settings are also saved with your project automatically.

> **TIP**
> Every control has a tooltip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

The orbit view fills the middle. The reverb's controls sit around it in glass cards, one per part of the sound.



1. **Preset bar.** Step with the arrows, click the name for the menu, **SAVE** your own.
2. **RUN / RESET.** Pause the orbits, or put the bodies back where they started.
3. **Clock.** FREE, SYNC or RETRIG, and the bars per lap when synced.
4. **Space card.** SPACE, SHAPE and ORBIT pages: the room, its shape and the voices.
5. **Orbit view.** The live simulation. Drag, throw, add and remove bodies here.
6. **Body readout.** The selected body's mass, speed, routing and voice.
7. **Voice strip.** Where each body's voice sits from left to right.
8. **Loop card.** LOOP, DRIVE and ECHO pages.
9. **Body cards.** Mass, velocity and a modulation target for each body.
10. **Add body.** Adds the next body, up to six.
11. **Output.** Width, Duck, Mix and Level, with stereo meters.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated, and most knobs also accept CLAP parameter modulation from the host. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
`Shift` **+ drag**	Fine adjustment.
Scroll wheel	Step the value up or down.
Double-click	Reset to the default value.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number.

Each knob draws two arcs. The inner arc is where you set it. When a body or the host is modulating the parameter, a **modulation ring** shows the value actually in use, moving in real time. The four knobs with an extended range carry a small tick at 100%.

Panel size

The panel is 1200 × 780 at 100%. Right-click any empty part of the orbit view and choose *UI Size* to scale it to 75%, 100%, 125%, 150% or 200%. The size you pick is remembered the next time the editor opens.

CHAPTER 05

How JVerb Works

A tank of sixteen delay lines, a loop around it, and a small solar system that decides where each part of the tail is heard.

Signal flow

Sound enters at the top left, is shaped and diffused, rings in the tank, and leaves through the body voices to the output. The hexagons are **slots**: the drive sits in exactly one of its three slots, and the echo in one of its three. Only the slot that is selected is active.

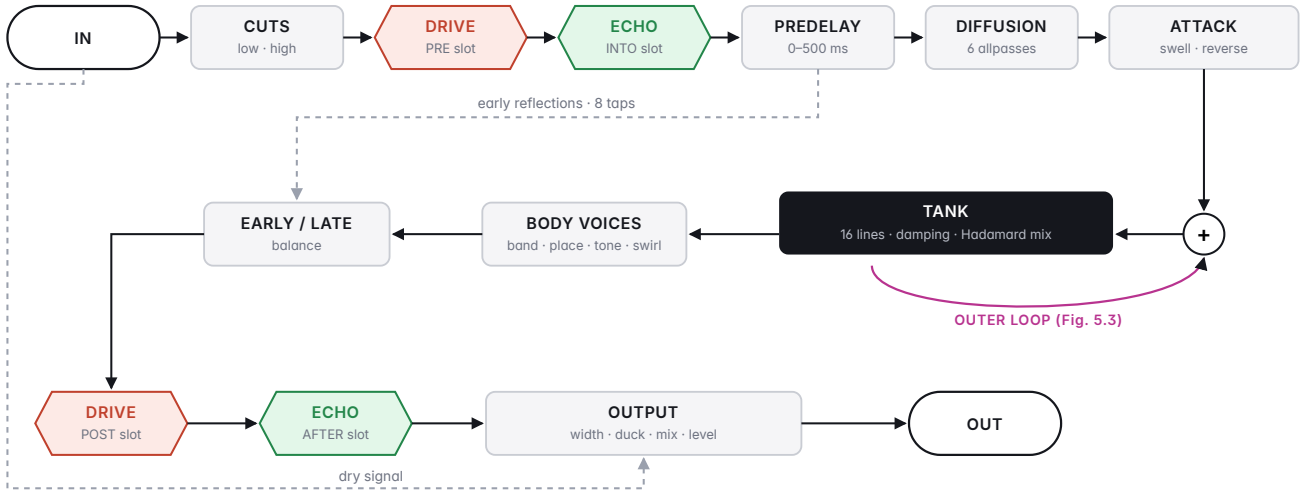


Fig. 5.1 The main signal path. Hexagons are slots for the drive (red) and echo (green). The dashed lines are the early reflections, which skip the tank, and the dry signal, which meets the wet at Mix.

The input stage

The input first passes **LOW CUT** and **HIGH CUT**. Only the reverb hears these filters; the dry signal is untouched. If the drive is in its PRE slot it comes next, then the echo if its slot is INTO. **PREDELAY** holds the sound back by up to half a second before the space answers.

Eight **early reflections** are tapped off the predelay, between 7 and 67 ms after it (scaled by System Size), alternating left and right. They never enter the tank: they go straight to the **EARLY/LATE** balance at the output. Meanwhile the signal is smeared by six allpasses per side (**DIFFUSION**) and shaped by **ATTACK**, which can soften it into a slow swell or play it backwards in overlapping windows, before it enters the tank.

The tank

The tank is a feedback delay network of sixteen lines. At System Size 1 they run from 29 ms to 202 ms, evenly spread on a log scale so that none is a simple ratio of another. Each pass through a line goes through two allpasses, loses a little level (so the tail falls 60 dB in the **DECAY** time), has its bass adjusted by **BASS MULT** and its treble damped by **DAMP HIGH**, and is then mixed into every other line through a 16 × 16 Hadamard matrix. The result is a sound that thickens into a dense wash within about 150 ms.

Each line also drifts very slowly on its own, which blurs resonances the way a chorus would. **MOTION** sets how far: at 100% no line bends a pass by more than 4 cents, so the tail never sounds pitchy.

Bodies are voices

The sixteen lines are dealt out among the bodies in left/right pairs, so every body owns a spread of short and long lines on both sides. With three bodies, Io gets lines 1–2, 7–8 and 13–14; Kepler gets 3–4, 9–10 and 15–16; Vesta gets 5–6 and 11–12. Each body's lines are summed into its own stereo **voice**, and the voice is shaped by where the body is:

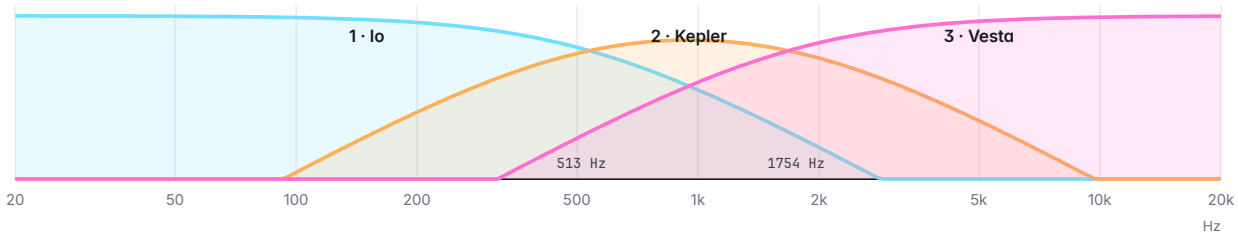


Fig. 5.2 With three bodies the spectrum splits into three log-spaced bands between 150 Hz and 6 kHz, body 1 lowest. The outer bands stay open below and above. At Orbit Pan 100% each voice is narrowed to its band, falling 12 dB per octave outside it.

VOICE	COMES FROM	CONTROLLED BY
Band	The body's number: body 1 lowest. With *n* bodies the spectrum is cut into *n* bands.	ORBIT · PAN
Place	The body's left/right position on the stage.	ORBIT · PAN
Tone	The body's distance from the centre of mass: bright near the centre, dark out on the swing.	ORBIT · TONE
Swirl	The body's angle around the centre. A four-stage phaser is swept twice per lap.	ORBIT · SWIRL

None of this bends pitch: the bodies never change a line's length. Because a body's tone also moves the damping in its own lines, a dark body's part of the tail dies darker as well as sounding darker.

The outer loop

Around the whole tank runs a second, slower loop. It takes the plain sum of all the lines, *before* the voices, shapes it, and feeds it back into the tank's input:

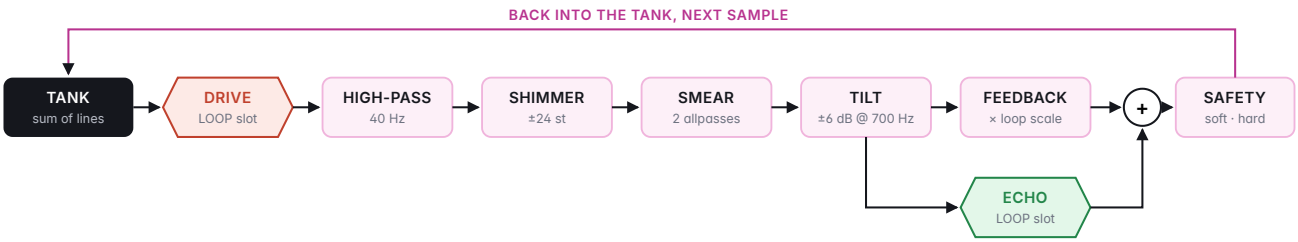


Fig. 5.3 The outer loop. Shimmer re-pitches every pass so the tail climbs (or falls); the safety stage keeps the loop bounded however hard it is pushed.

Because the loop is taken before the voices, nothing a voice does (a boost, a ringing phaser, a stereo flip) is ever fed back. That is what makes the voice controls safe to push to 200%.

FEEDBACK is scaled so that it means the same thing at any decay and size: below roughly 100% the tail still dies away, just longer and more coloured by the loop; past it, the loop grows until the safety stage holds it.

Bodies are modulators too

Besides its voice, each body produces a **modulation signal**: how fast it is moving compared with the others, plus how close its nearest neighbour is. A body cruising at the average speed sits near zero; a body slingshotting past another spikes high. Route that signal to one of nine parameters with the body's **Target** and **Amount** (Chapter 9).

The orbit simulation

The bodies obey Newtonian gravity: every body pulls on every other in proportion to its mass. **GRAVITY** scales the pull, so higher values give faster, tighter orbits with the same shapes. The simulation takes smaller steps during close passes so slingshots stay accurate, keeps the system centred, and gently catches any body flung too far out and turns it around. Simulation and audio run on the same 32-sample clock, so the same settings and input always give the same orbits and the same render, at any buffer size.

CHAPTER 06

The Orbit View

The stage in the middle of the panel is not just a display. It is an instrument: every body can be grabbed, thrown and weighed.

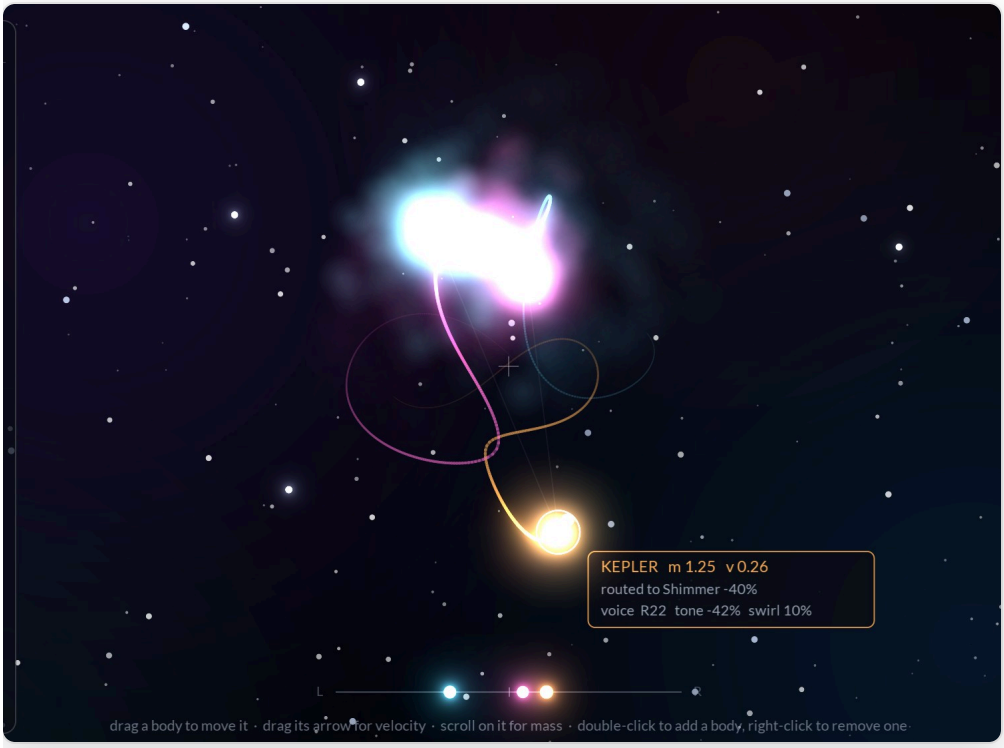


Fig. 6.1 Three bodies on the stage. Kepler is selected: its readout shows its mass, speed, routing and voice. The strip at the bottom places each voice from left to right.

Gestures

GESTURE	WHERE	WHAT IT DOES
Drag	On a body	Moves the body. Release it and it carries on from where you left it.
Drag the arrow	On the selected body	Sets the body's velocity: direction and speed.
Scroll	On a body	Changes its mass. Heavier bodies are drawn bigger.
Double-click	Empty space	Adds a body at that point, moving at circular speed around the others (up to six).
Right-click	On a body	Removes it. The bodies after it move down a place, settings and all.
Right-click	Empty space	The panel menu, with *UI Size*.
Click a card	Body cards	Selects that body; click again to deselect.

The bodies

Six bodies are available, each with a fixed name and colour. The first three are present by default.

NO.	BODY	DEFAULT MASS	FOLLOWS TRANSIENTS IN
1	**Io**	1.00	Lows (below 250 Hz)
2	**Kepler**	1.25	Mids
3	**Vesta**	0.80	Highs (above 2.5 kHz)
4	**Titan**	1.00	Lows
5	**Rhea**	1.00	Mids
6	**Luna**	1.00	Highs

Three bodies start on the figure-eight orbit, which is stable when the masses are equal. JVerb's default masses (1, 1.25, 0.8) break it up within a few laps into motion that never repeats. Any other number of bodies starts on a ring with slightly different radii, so the ring soon breaks up too.

Reading the view

Everything on the stage reflects what you hear:

- **Glow.** A body whose voice is bright burns whiter and wider; a dark one burns deeper in its own colour.
- **Arcs.** Arcs spin around a body faster the harder its phaser is sweeping.
- **Voice strip.** The line under the stage, marked L and R, places each voice in the stereo field.
- **Gas.** Each transient in the input blows a shell of soft gas out of the bodies that follow its band. It billows, drifts on a slow swirl and falls back. Where two bodies' clouds meet they push each other apart and the seam glows whiter.
- **Dust.** Each band also makes its bodies glow and throw off sparks of dust, which fall through the system's gravity.
- **AU rings.** The system is drawn at its System Size: a quarter of the stage at 0.25 AU, all of it at 10 AU. Faint rings mark the scale while you turn the knob or hover over the system.
- **Readout.** The selected body's box shows its mass (m) and speed (v), where it is routed and by how much right now, and its voice: position, tone and swirl.

> **REACT**
> Transients do more than draw gas. With **REACT** (on the SHAPE page) above zero, each hit also kicks the bodies that follow its band, so a busy drum part keeps reshaping the orbit, and through it the sound. Set React to 0% for orbits that only follow gravity.

CHAPTER 07

Presets and Clock

The header holds the preset browser and the controls that run the orbits.



Fig. 7.1 The header: preset bar, RUN and RESET, the clock switch and Sync Bars.

The preset bar

- ◀ ▶ step to the previous or next preset, in menu order. Scrolling over the bar does the same.
- **The name** opens the preset menu: the factory bank in four categories (*Short*, *Medium*, *Long*, *Washes*), a *User* submenu with your own presets, and *Save as…*. The category of the current preset is shown at the left of the bar.
- **SAVE** asks for a name and writes the current settings as a user preset.

Loading a preset sets every parameter, so nothing carries over from the previous sound (a body's routing, for example). The host sees the change as an ordinary edit, so it can be undone. The orbits then restart from the preset's masses and velocities.

User presets

User presets are small text files with the extension `.jvpreset`, one line per parameter, so they sound the same even if a later version changes a default. They live in:

macOS	`~/Library/Application Support/JVerb/Presets`
Windows	`%APPDATA%\JVerb\Presets`
Linux	`$XDG_CONFIG_HOME/JVerb/Presets` (or `~/.config/JVerb/Presets`)

Copy files into this folder to share presets between machines; they appear the next time the menu opens. The name of the last preset loaded or saved is stored with your project.

Running the orbits

CONTROL	RANGE	DEFAULT	WHAT IT DOES
RUN	`On / Off`	`On`	Off stops the orbits where they are. The tail keeps sounding, with every voice held where its body stopped.
RESET	`button`	–	Puts the bodies back where they start, with their set masses and velocities.
Clock	`Free, Sync, Retrig`	`Free`	How the orbits are clocked (below).

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Sync Bars	`1 to 16 bars`	`4`	Bars per lap of the start orbit when the clock is SYNC.

Clock modes

FREE The orbits run on their own. With three bodies the figure-eight comes round about every 8 seconds at Gravity 1.

SYNC One lap of the start orbit fits into **Sync Bars** bars at the host's tempo and time signature. When the transport starts, loops or jumps, the orbits restart and run on to where they would be at that point in the cycle, so they line up with your bars.

RETRIG The orbits run free, but restart whenever the transport starts, loops or jumps.

Restarts are deterministic: the same settings and the same input give the same orbits, and the same render, every time. That makes SYNC and RETRIG the modes to use when you bounce or freeze tracks.

CHAPTER 08

The Space Card

Three pages on the left of the panel: SPACE sets the room, SHAPE its character, ORBIT how much the bodies move their voices.

PAGE 1

SPACE



CONTROL	RANGE · DEFAULT	WHAT IT DOES
DECAY	0.2 – 60 s **4.00 s**	How long the tail lasts: the time for the mids to fall 60 dB (RT60). MOD TARGET
SYSTEM SIZE	0.25 – 10 AU **1.00 AU**	Scales every line in the tank, the early reflections, and the swell and reverse windows. At 10 AU the longest line is 2 s. Also sets how big the stage draws the system. MOD
GRAVITY	G ×0.25 – ×4 **×1.00**	How hard the bodies pull. Higher runs the orbits faster and tighter with the same shapes. MOD
BODIES	2 – 6 **3**	How many bodies, and so how many voices and bands. Fewer is sparse, more is dense.
MOTION	0 – 200% **40%**	How far the tank's lines drift: a chorus-like blur up to 100%, warble beyond. MOD
PREDELAY	0 – 500 ms **20 ms**	Delay before the space answers. MOD
DAMP HIGH	1 – 20 kHz **7.0 kHz**	Highs above this die faster on every pass, so the tail darkens as it rings. MOD TARGET

SYSTEM SIZE AND DECAY ARE INDEPENDENT

Decay is always the RT60 you set, whatever the size. A small system with a long decay gives a dense, metallic ring; a big system with a short decay sounds like a large space heard briefly, with more audible early structure. Sizes above about 4 AU start to sound grainy and echo-like by design.

PAGE 2

SHAPE



CONTROL	RANGE · DEFAULT	WHAT IT DOES
ATTACK	Reverse 100% – Swell 100% **Off**	Right of centre the tail swells in through three long allpasses; left of centre it plays the input backwards in overlapping windows (80 to 580 ms). MOD
DIFFUSION	0 – 100% **70%**	How much the input is smeared before the tank, and how dense the tank's own allpasses are. At 0% a click goes in as a click; at 100% as a cloud. MOD TARGET
EARLY/LATE	0 – 100% late **75% late**	Equal-power balance of early reflections and tail. MOD TARGET
BASS MULT	×0.25 – ×2.5 **×1.20**	Bass decay time relative to Decay. MOD
BASS XOVER	80 – 800 Hz **250 Hz**	Where Bass Mult takes over. MOD
LOW CUT	20 Hz – 1 kHz **40 Hz**	High-pass on the reverb's input. The dry signal is untouched. MOD
HIGH CUT	1 – 20 kHz **16.0 kHz**	Low-pass on the reverb's input. MOD
REACT	0 – 100% **30%**	How hard transients in the input kick the bodies. MOD

Swell and reverse

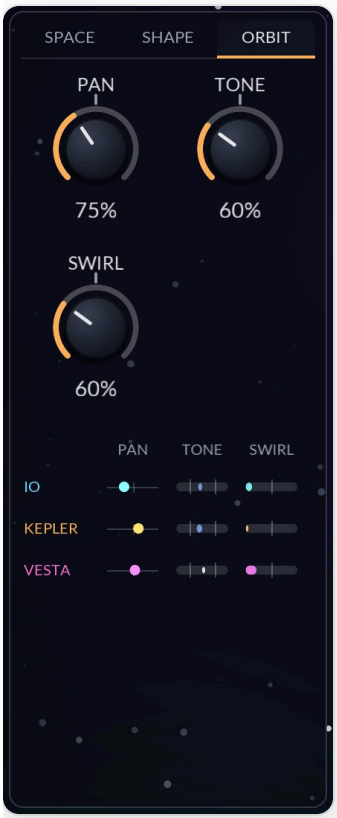
ATTACK is a bipolar knob. Turned right, transients soften into a slow bloom; the swell's allpasses grow with System Size. Turned left, the diffused signal is played backwards in two overlapping windows that grow from 80 ms to 580 ms as you turn, giving the classic reverse-reverb suck-in on every note. Both crossfade in gradually, so small settings just take the edge off.

React in detail

A three-band analyser (split at 250 Hz and 2.5 kHz) watches the dry input for sudden jumps. Each transient kicks the bodies following its band, in proportion to React and how hard it hit. Kicks also scale with Gravity, so they feel the same at any orbit speed.

PAGE 3

ORBIT



CONTROL	RANGE · DEFAULT	WHAT IT DOES
PAN	0 – 200% 75%	Narrows each voice to its own band and moves it from a wide spread to where its body is, left to right. Past 100% voices are thrown to the edges and their stereo turns inside out. MOD
TONE	0 – 200% 60%	A voice darkens as its body swings out and brightens near the centre. Also moves the damping in that body's lines. MOD
SWIRL	0 – 200% 60%	How deep each voice's four-stage phaser goes, swept twice per lap by its orbit. Past 100% the phasers feed back until they ring. MOD

THE VOICE LIST

Below the knobs, a row per body shows its voice live: where it sits from left to right, its tone (dark to bright), and how hard it is swirling. Each meter draws the same 0 to 200% scale as the knobs, with a tick at 100%.

How tone follows distance

Each body's tone depends on its distance from the centre of mass, scaled by **TONE**. Near the centre a voice is bright, out on the swing it is dark, and at about 0.9 units it is neutral.

BODY'S TONE	ABOVE 2.5 KHZ	LEVEL	DAMPING IN ITS LINES
Dark (–1)	−12 dB	−3 dB	an octave lower
Bright (+1)	+4 dB	+3 dB	an octave higher
Dark, past 100% (–2)	−24 dB	−3 dB	three octaves lower
Bright, past 100% (+2)	+14 dB	+3 dB	three octaves higher

Mass and velocity shape this too: a heavy body sits near the centre and stays bright, while its light partners swing wide and go dark and wide.

With Pan, Tone and Swirl all at 0%, every voice passes its lines straight through, and JVerb behaves like a conventional, very smooth stereo reverb.

CHAPTER 09

Body Cards and Modulation

Each body has a card along the bottom of the panel: two knobs that change its orbit, and a routing that lets it drive the sound.



Fig. 9.1 Body cards for Io, Kepler (selected, routed to Shimmer) and Vesta (routed to Tilt), and the ADD BODY slot. With more bodies the cards narrow and VELOCITY and AMOUNT are labelled SPEED and AMT.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MASS	`0.10 – 3.00`	`1.00`, `1.25`, `0.80`, then `1.00`	The body's gravity on the others, and how big it is drawn. A heavy body sits near the centre; light ones swing wide around it. MOD
VELOCITY	`×0.50 – ×1.50`	`×1.00`	Scales the body's start speed. Turned while running, it scales the body's current speed by as much. Slow bodies fall inward; fast ones fly wide. MOD
Target	`menu`	`Off`	Which parameter this body's modulation signal moves.
AMOUNT	`−100% – +100%`	`+50%`	How much, and in which direction. MOD

Click **ADD BODY** to add the next body, up to six, or use the **BODIES** stepper on the SPACE page. A new body enters outside the others, moving at circular speed around them. Lowering the count removes the newest body.

The modulation signal

Each body's signal combines two things: how fast it is moving compared with the average of all the bodies, and how close its nearest neighbour is. The result runs from –1 to +1 and is smoothed over about 60 ms. A body slingshotting past another spikes high; a body crawling at the far end of a long orbit goes negative. The selected body's readout shows the live value of its routing.

Targets

Bodies start routed to *Off*, so nothing moves the sound that the panel doesn't show. When a body is routed, its target knob's modulation ring moves with it. Several bodies on the same target add up, and each result is kept within its parameter's range. Host modulation is added first, then the bodies' offset.

TARGET	AT FULL SIGNAL × 100% AMOUNT	TARGET	AT FULL SIGNAL × 100% AMOUNT
Decay	Decay ×5.7 (÷5.7 at –1)	**Shimmer**	+125 points
Damping	Damp High moves 3.75 octaves	**Feedback**	+100 points
Diffusion	+125 points	**Tilt**	+200 points
Early/Late	+125 points	**Width**	+200 points
Drive	+45 dB	**Off**	No modulation

> **SOUND-DESIGN IDEA**
> Route a light, fast body to **Damping** with a negative Amount: each slingshot briefly darkens the tail, like a door swinging shut. Route a heavy, slow body to **Width** for a gentle breathing of the stereo image. Because the orbit never quite repeats, neither does the modulation.

CHAPTER 10

The Loop Card

Three pages on the right: the outer loop around the tank, the drive, and the echo.

PAGE 1

LOOP



CONTROL	RANGE · DEFAULT	WHAT IT DOES
FREEZE	`On / Off` **`Off`**	Holds the tail forever, at its level and brightness, and lets nothing new in.
FEEDBACK	`0 – 150%` **`0%`**	Sends the tank's output back to its input through the loop. Scaled to mean the same at any decay. MOD TARGET
SHIMMER	`0 – 100%` **`0%`**	How much of the loop is pitch shifted. Heard even with Feedback at 0. MOD TARGET
PITCH	`−24 – +24 st` **`+12 st`**	The shimmer interval, in semitones.
TILT	`−100% – +100%` **`+0%`**	Darkens or brightens what goes round the loop: up to 6 dB each way around 700 Hz. Below 40% Feedback it also tilts the tail you hear, up to 9 dB each way. MOD TARGET
SMEAR	`0 – 100%` **`30%`**	Allpass blur in the loop. MOD
SAFETY	`Soft, Hard` **`Soft`**	How the loop is kept from running away: a soft saturator or a hard clip.

Feedback, and past 100%

With Feedback below roughly 100% the tail still dies away, just longer and more coloured by the loop's shimmer, smear and tilt. Around 100% (somewhere between about 96% and 123%, depending on Decay and Size) the tail starts to sustain itself; past that, it grows until the **SAFETY** stage holds it, and FEEDBACK runs on to 150% so every Decay and Size has room to scream. *Soft* rounds the peaks off with a gentle saturation; *Hard* clips them for a harsher, more aggressive sustain.

Shimmer

Each pass through the loop is pitch shifted by **PITCH**, and Shimmer sets how much of the loop takes the shifted path. Because every pass is shifted again, the tail climbs (or descends) in steps: +12 gives the classic octave-up halo, –12 a deepening undertow, and a fifth (+7) a harmonic stack. Combine with

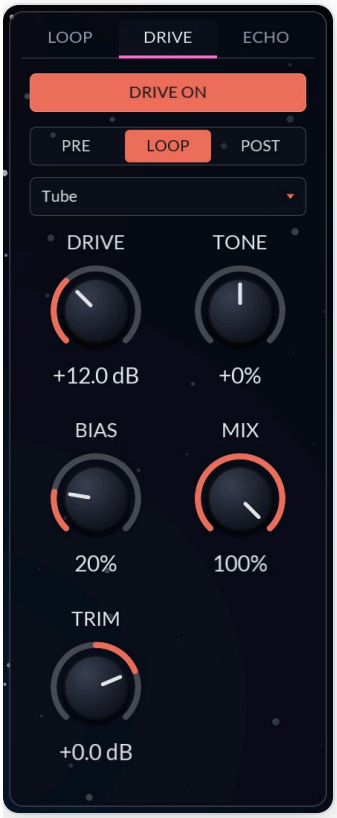
Feedback for long, building shimmers.

Freeze

FREEZE sets every line to sustain exactly, fades the damping out, slows the drift to a stop and closes both the input and the loop. Whatever is ringing at that moment holds indefinitely. Play over it with the dry signal, then release Freeze to let it decay naturally.

PAGE 2

DRIVE



CONTROL	RANGE · DEFAULT	WHAT IT DOES
DRIVE ON	`On / Off` `Off`	Switches the drive in. Fades over a few milliseconds rather than clicking.
Slot	`PRE, LOOP, POST` `PRE`	PRE drives the input; LOOP drives every pass of the feedback loop; POST drives the reverb's output.
Type	`menu` `Tube`	Tape, Tube, Fuzz, Fold or Crush (below).
DRIVE	`0 – +36 dB` `+12 dB`	Gain into the shaper. MOD TARGET
TONE	`-100% – +100%` `+0%`	Tilts what goes into the shaper around 900 Hz, and back after, so it changes *what* distorts, not the overall balance. MOD
BIAS	`0 – 100%` `20%`	Asymmetry for even harmonics (Tube, Fuzz), fold depth (Fold), or hold length (Crush). MOD
MIX	`0 – 100%` `100%`	Clean / driven blend. MOD
TRIM	`-18 – +6 dB` `0 dB`	Output trim. In the LOOP slot it can only cut. MOD

Drive types

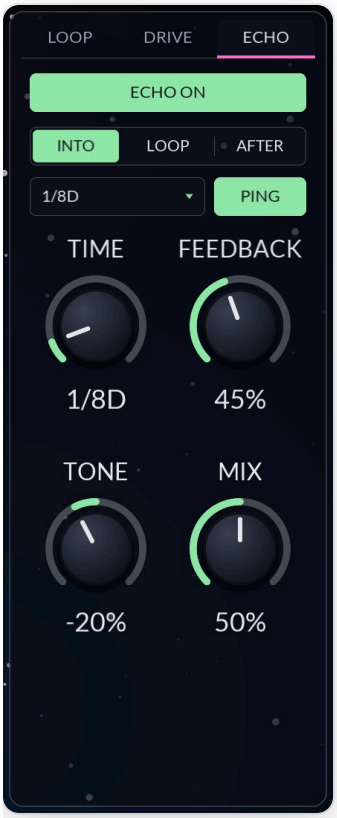
TYPE	CHARACTER	WHAT BIAS DOES
Tape	Smooth, symmetrical saturation. Warm and forgiving.	–
Tube	Asymmetric saturation with more even harmonics.	More asymmetry, more even harmonics.
Fuzz	A hard exponential knee. Buzzy and aggressive.	Offsets the knee.
Fold	A sine wavefolder that folds back on itself as it is driven. Metallic, synth-like.	Fold depth.
Crush	Bit reduction from 12 down to 3 bits as Drive rises.	Sample-and-hold: holds each sample for up to 15 samples.

Tape, Tube, Fuzz and Fold run at 2× oversampling to keep aliasing down. Crush is deliberately not oversampled.

> **DRIVE IN THE LOOP**
> In the LOOP slot the drive works on every pass, so the tail gets dirtier the longer it rings. JVerb makes sure a quiet signal never comes out of the loop drive louder than it went in, so adding drive never makes the loop run away below 100% Feedback. That is also why TRIM can only cut in this slot. Past 100% Feedback the drive is let through more and more, until at 150% none of its gain is taken back: the loop then holds and screams instead of choking.

PAGE 3

ECHO



CONTROL	RANGE · DEFAULT	WHAT IT DOES
ECHO ON	`On / Off` **`Off`**	Turns the echo's output up or down. The echo keeps running underneath, so switching on never replays old repeats.
Slot	`INTO, LOOP, AFTER` **`INTO`**	Where the echo sits (below).
Sync	`Free, 1/16 … 2 Bars` **`1/8D`**	A note value at the host tempo, or Free to use TIME.
PING	`On / Off` **`On`**	Ping-pong: the first repeat is on the left, then they alternate.
TIME	`10 – 4000 ms` **`375 ms`**	The echo time when Sync is Free. Otherwise the knob shows the note value. MOD
FEEDBACK	`0 – 105%` **`45%`**	Level of each repeat relative to the last. Past 100% the repeats saturate rather than run away. MOD
TONE	`−100% – +100%` **`−20%`**	Darker or brighter with every repeat. MOD
MIX	`0 – 100%` **`50%`**	Level of the repeats added at the slot. MOD

The three echo slots

Wherever it sits, the echo works as a send and return: it takes a copy of the signal at that point and adds only its repeats back, so the signal itself always passes straight through.

SLOT	LISTENS TO	WHAT YOU HEAR
INTO	The input, after the cuts and Drive PRE.	Every repeat is reverberated: the space hears each one as a new note, giving a rhythm of reverb swells.
LOOP	The outer loop, after shimmer, smear and tilt.	The tail comes back into the tank in long, spaced passes, one per echo time, even with Feedback at 0. With Feedback up, the two loops stack.

SLOT	LISTENS TO	WHAT YOU HEAR
AFTER	The finished wet signal, after Drive POST.	Plain echoes of the reverb. They are part of the wet, so Width, Duck and Mix act on them, but they are never reverberated.

Time and sync

Sync values run 1/16, 1/8T, 1/16D, 1/8, 1/4T, 1/8D, 1/4, 1/2T, 1/4D, 1/2, 1/2D, 1 Bar and 2 Bars; the bar values follow the host's time signature. At 120 BPM the default 1/8D is 375 ms. A synced time longer than 4 s, at slow tempos, is halved until it fits.

Changing the time bends the repeats already in the line, like a tape machine settling to a new speed, rather than clicking.

Echo tone

Tone works on the feedback only, so each later repeat is shaped more than the one before.

TONE	PER PASS, LOW	PER PASS, HIGH
−100%	0 dB	−12 dB
−20%	0 dB	−2.4 dB
0%	0 dB	0 dB
+100%	−9 dB	+3 dB

Split at 1.8 kHz. A 60 Hz high-pass in the feedback keeps lows from piling up.

SWITCHING WITHOUT CLICKS

Turning the drive on or off, changing its slot or type, or moving the echo to another slot all fade over a few milliseconds instead of switching in one sample. The echo keeps what is already in its lines when it moves, so you can change slots mid-performance.

CHAPTER 11

Output

The last stage: stereo width, ducking, the dry/wet blend and the final level.



Fig. 11.1 The Output card with its stereo peak meters.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
WIDTH	`0 – 200%`	`100%`	Stereo width of the wet signal: 0% mono, 100% as is, 200% doubled side. MOD TARGET
DUCK	`0 – 100%`	`0%`	Turns the reverb down by up to this much while the input plays (5 ms attack, 300 ms release), so it blooms in the gaps. MOD
MIX	`0 – 100%`	`35%`	Dry/wet balance. Use 100% on a send. MOD
LEVEL	`−24 – +12 dB`	`+0.0 dB`	Output gain, on dry and wet together. MOD

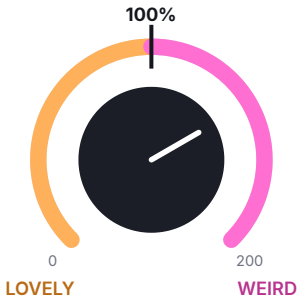
The meters show the peak level of each output channel. With high Feedback, Shimmer or Drive the wet level can rise well above the input: keep an eye on them, and use **LEVEL** or the loop's **SAFETY** mode to tame it.

JVerb also reports its tail length to the host, worked out from the current Decay, Predelay, Feedback and echo settings, so hosts that stop processing silent tracks let the reverb ring out fully. With Freeze on, Feedback at 95% or more, or the echo in its LOOP slot, the tail is reported as infinite.

CHAPTER 12

The Extreme Zone

Four knobs go to 200%. Past the tick at 100% is where the weird stuff lives.



Up to the tick the four knobs are tuned to stay lovely: nothing bends pitch audibly and nothing rings.

Past the tick they keep going, to twice their normal range. The ORBIT page's voice list draws the same 0 to 200% scale with the same tick, so you can see which voices are in the extreme zone.

KNOB	0 – 100%	100 – 200%
MOTION	The lines drift up to 4 cents a pass. A chorus-like blur, never pitchy.	The drift grows to as much as 150 cents a pass and runs up to four times faster. With Feedback high the tail warbles and groans.
ORBIT PAN	Voices narrow to their bands and move from a wide spread to where their bodies are.	Voices are thrown further out (at 200%, six times as far), so most pin to the edges, and each voice's stereo turns inside out.
ORBIT TONE	Up to –12 / +4 dB above 2.5 kHz, ±3 dB of level, damping ±1 octave.	Up to –24 / +14 dB above 2.5 kHz and damping ±3 octaves, so the swing between bodies gets hard.
ORBIT SWIRL	Phaser depth up to its full mix; two sweeps per lap.	The phasers feed back on themselves until they ring, the sweep widens and goes round up to eight times per lap, and the tank's own allpasses swirl twice as hard.

Only Motion acts inside the tank. Because the outer loop is taken before the voices, nothing Pan, Tone or Swirl does is ever fed back, so even at 200% they cannot make the reverb run away. The ringing of an extreme Swirl is level-compensated so its loudest resonance stays at unity.

Three other parameters also go past unity: **FEEDBACK** to 150% (the tail grows into the safety stage), echo **FEEDBACK** to 105% (repeats saturate), and **WIDTH** to 200%.

CHAPTER 13

Factory Presets

Thirty-two starting points, ordered from tight rooms to washes that ring for most of a minute.

7 PRESETS

Short

Rooms, booths and slapbacks under about a second.

Init	The defaults: a 4 s tail with three bodies.
Tight Booth	0.35 s, the smallest size, mostly early reflections.
Wood Room	0.7 s, small, a little dark, lighter bass.
Small Orbit	1 s, four bodies, bright.
Drum Reactor	React 100%, Duck and Low Cut; Io routed to Damping.
Swirling Slapback	A 140 ms straight echo INTO the reverb, free-running; Io on Damping.
Hot Input	Tape drive in the PRE slot at +15 dB.

10 PRESETS

Medium

Plates, halls and character spaces.

Figure-Eight Plate	2.4 s, Diffusion 95%, all late, bright.
Ping-Pong Space	Synced 1/4 ping-pong echo AFTER the reverb.
Vocal Hall	2.8 s, 35 ms predelay, Low Cut, Duck.
Dotted Echo Hall	Synced 1/8D echo INTO a 3 s hall.
Binary Star	Two bodies, a heavy pair; Kepler on Width.
Synced Orbit	Clock SYNC, 2 bars a lap; Vesta on Tilt.
Burning Loop	Tube drive in the LOOP at +18 dB, Feedback 55%, dark tilt.
Crushed Satellites	Crush in the LOOP, five bodies, Feedback 40%.
Chaos Engine	Six bodies, Gravity 2.5, React 80%; bodies on Damping and Feedback.
Heavy Moon	One heavy body (2.8) and two light ones.

9 PRESETS

Long

Long tails, loops and shimmers.

Tape Loop Orbit	A 1-bar echo in the LOOP plus tape drive in the LOOP; Vesta on Tilt.
Swell Machine	Swell 85%, 6 s.
Fuzz Cloud	Fuzz POST at +24 dB, 60% mix, 7 s.
Folded Space	Fold in the LOOP, Shimmer 25%, Feedback 80%.
Slow Ballet	Gravity 0.35, Smear 60%, 8 s.
Shimmer Halo	Octave-up shimmer 45%, Feedback 85%, 9 s.
Infinite Return	A 1/2-note echo in the LOOP, Feedback 70%, a little shimmer.
Octave Under	Octave-down shimmer, Feedback 80%, dark.
Cathedral Drift	13 s, Size 2.6, 45 ms predelay, long bass.

6 PRESETS

Washes

Ambient beds that ring for most of a minute.

Nebula	22 s, Size 3, slow attack, wide.
Deep Field	30 s, six bodies, low gravity, 60 ms predelay.
Galaxy Shimmer	35 s, octave-up shimmer, Feedback 50%, wide.
Event Horizon	45 s, Size 4, Diffusion 100%, dark, 80 ms predelay.
Dark Matter	50 s, Size 4, very dark, octave-down shimmer, long bass.
Frozen Ring	60 s, all late, a little Duck.

> **ECHO SLOTS IN THE FACTORY BANK**
> To hear each echo slot: *Swirling Slapback* and *Dotted Echo Hall* use INTO, *Tape Loop Orbit* and *Infinite Return* use LOOP, and *Ping-Pong Space* uses AFTER.

CHAPTER 14

Sound Design Recipes

Starting points for common jobs and uncommon sounds. Begin from *Init* unless a preset is named, then change only what is listed.

Vocal plate that stays out of the way

A bright, dense tail that ducks under the singer and blooms between phrases.

```
DECAY 2.8 s · PREDELAY 35 ms
DIFFUSION 95% · EARLY/LATE 100%
LOW CUT 150 Hz · DUCK 40%
```

Close to: Vocal Hall, Figure-Eight Plate

Drum room that answers back

The kit kicks the bodies around; one body darkens the room on every slingshot.

```
DECAY 1.0 s · REACT 100% · DUCK 30%
LOW CUT 120 Hz
IO → Damping, AMOUNT −20%
```

Close to: Drum Reactor

Rhythmic bloom

Each dotted-eighth repeat swells into its own reverb.

```
ECHO ON · slot INTO · SYNC 1/8D
ECHO FEEDBACK 45% · ECHO MIX 60%
DECAY 3 s
```

Close to: Dotted Echo Hall

Octave halo

A climbing, angelic shimmer for pads and guitars.

```
SHIMMER 45% · PITCH +12 st
FEEDBACK 85% · DECAY 9 s
TILT −20% · MIX 50%
```

Close to: Shimmer Halo

Tape-loop ambience

The tail returns once a bar, a little dirtier each time.

```
ECHO ON · slot LOOP · SYNC 1 Bar
DRIVE ON · slot LOOP · type Tape
VESTA → Tilt
```

Close to: Tape Loop Orbit

Endless drone bed

Catch a chord in the tail and hold it while you play over the top.

```
DECAY 30 s · SIZE 3 AU · ATTACK Swell 50%
Play a chord, then FREEZE On
MIX 100% on a send
```

Close to: Nebula, Frozen Ring

Locked to the groove

The orbit repeats every two bars and lines up with the song on every playback.

```
Clock SYNC · Sync Bars 2
ORBIT PAN 120% · SWIRL 80%
VESTA → Tilt, AMOUNT +24%
```

Close to: Synced Orbit

Seasick space

Into the extreme zone: a warbling, groaning tail that turns inside out.

```
MOTION 160% · FEEDBACK 95%
ORBIT PAN 200% · SWIRL 170%
DECAY 12 s · SAFETY Soft
```

Pull Motion back below the tick to tame it.

General tips

- **Fewer bodies, clearer voices.** Two bodies give two wide bands that swap places; six give a busy, finely sliced shimmer of motion.
- **Mass sets the drama.** One heavy body with light partners gives a stable centre and wild outer voices (*Heavy Moon*). Equal masses give the most even, flowing motion.
- **Gravity sets the tempo of the swirl.** Low Gravity is a slow ballet; high Gravity a frantic, tight cluster. For tempo-locked motion use SYNC instead.
- **Pause to audition.** Turn RUN off to freeze the voices where they are while you adjust the reverb itself.
- **Long washes on a send.** Decays over 20 seconds sit best on a send with Mix at 100%, Low Cut raised and a little Duck.

APPENDIX A

Parameter Reference

Every parameter JVerb exposes to the host, in host order. Everything is automatable.

SPACE

PARAMETER	RANGE	DEFAULT	
Decay	0.2 – 60 s	4.00 s	MOD TARGET
System Size	0.25 – 10 AU	1.00 AU	MOD
Gravity	G ×0.25 – ×4	×1.00	MOD
Bodies	2 – 6	3	
Motion	0 – 200%	40%	MOD
Attack	Reverse 100% – Swell 100%	Off	MOD
Diffusion	0 – 100%	70%	MOD TARGET
Early/Late	0 – 100% late	75% late	MOD TARGET
Predelay	0 – 500 ms	20 ms	MOD
Damp High	1 – 20 kHz	7.0 kHz	MOD TARGET
Bass Mult	×0.25 – ×2.5	×1.20	MOD
Bass Xover	80 – 800 Hz	250 Hz	MOD
Low Cut	20 Hz – 1 kHz	40 Hz	MOD
High Cut	1 – 20 kHz	16.0 kHz	MOD

ORBIT

PARAMETER	RANGE	DEFAULT	
Run	On / Off	On	
Clock	Free, Sync, Retrig	Free	
Sync Bars	1 – 16	4	
React	0 – 100%	30%	MOD
Orbit Pan	0 – 200%	75%	MOD
Orbit Tone	0 – 200%	60%	MOD
Orbit Swirl	0 – 200%	60%	MOD

BODIES (PER BODY: IO, KEPLER, VESTA, TITAN, RHEA, LUNA)

PARAMETER	RANGE	DEFAULT	
Mass	0.10 – 3.00	1.00 (Kepler 1.25, Vesta 0.80)	MOD
Velocity	×0.50 – ×1.50	×1.00	MOD
Target	Off, Decay, Damping, Diffusion, Early/Late, Drive, Shimmer, Feedback, Tilt, Width	Off	
Amount	-100% – +100%	+50%	MOD

LOOP

PARAMETER	RANGE	DEFAULT	
Feedback	0 – 150%	0%	MOD TARGET
Shimmer	0 – 100%	0%	MOD TARGET
Shimmer Pitch	-24 – +24 st	+12 st	
Tilt	-100% – +100%	+0%	MOD TARGET
Smear	0 – 100%	30%	MOD
Safety	Soft, Hard	Soft	
Freeze	On / Off	Off	

DRIVE

PARAMETER	RANGE	DEFAULT	
Drive On	On / Off	Off	
Drive Slot	Pre, Loop, Post	Pre	
Drive Type	Tape, Tube, Fuzz, Fold, Crush	Tube	
Drive	0 – +36 dB	+12 dB	MOD TARGET
Drive Tone	-100% – +100%	+0%	MOD
Drive Bias	0 – 100%	20%	MOD
Drive Mix	0 – 100%	100%	MOD
Drive Trim	-18 – +6 dB	0 dB	MOD

ECHO

PARAMETER	RANGE	DEFAULT	
Echo On	On / Off	Off	
Echo Slot	Into, Loop, After	Into	
Echo Time	10 – 4000 ms	375 ms	MOD
Echo Sync	Free, 1/16, 1/8T, 1/16D, 1/8, 1/4T, 1/8D, 1/4, 1/2T, 1/4D, 1/2, 1/2D, 1 Bar, 2 Bars	1/8D	
Echo Feedback	0 – 105%	45%	MOD

PARAMETER	RANGE	DEFAULT	
Echo Tone	-100% – +100%	-20%	MOD
Echo Ping-Pong	On / Off	On	
Echo Mix	0 – 100%	50%	MOD

OUTPUT

PARAMETER	RANGE	DEFAULT	
Width	0 – 200%	100%	MOD TARGET
Duck	0 – 100%	0%	MOD
Mix	0 – 100%	35%	MOD
Level	-24 – +12 dB	0 dB	MOD

APPENDIX B

Specifications

Format	CLAP 1.2 audio effect. Features: audio-effect, reverb, stereo.
Channels	Stereo in, stereo out. A mono input is used on both sides.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit).
Latency	None reported to the host.
Tail	Reported to the host from the current settings, so reverb tails are never cut off.
Reverb tank	16-line feedback delay network, two allpasses per line, 16 × 16 Hadamard mix. Lines 29–202 ms at Size 1, up to 2.02 s at Size 10.
Decay	0.2 to 60 s RT60, with separate bass multiplier and crossover.
Early reflections	8 taps, 7 to 67 ms × System Size (up to 4×).
Simulation	2 to 6 bodies under mutual gravity, velocity Verlet with adaptive substeps.
Control rate	Every 32 samples; gains ramp across each block, so nothing zippers. Renders are identical at any buffer size.
Drive	5 shapes, 2× oversampled (31-tap half-band filters), except Crush.
Echo	Stereo, 10 ms to 4 s, 13 tempo-synced values plus Free, ping-pong.
Parameters	All automatable; most accept CLAP parameter modulation.
State	Saved with the host project, including the preset name. Older JVerb states load, with any missing parameters at their defaults.
Panel	1200 × 780 at 100%; 75%, 100%, 125%, 150% and 200%.
Threading	The audio thread never allocates, locks or waits on the panel after activation.

Troubleshooting

SYMPTOM	WHAT TO CHECK
JVerb doesn't appear in my host (macOS)	The download is probably quarantined. See the warning in Chapter 2, then rescan plug-ins.
The tail never stops	Check **FREEZE**, and whether **FEEDBACK** is near or above 100%. A body routed to Feedback or Decay can also push it over. Echo Feedback above 100% sustains the repeats.
The tail sounds out of tune or seasick	**MOTION** is past 100%, or **SHIMMER** is up with an interval you didn't intend.
A whistling or ringing tone	**ORBIT SWIRL** past 100% makes the phasers ring. A small System Size with a long Decay can also sound metallic.

SYMPTOM	WHAT TO CHECK
The reverb jumps around the stereo field too much	Lower **ORBIT PAN**, reduce **REACT**, or lower **GRAVITY** for slower motion.
SYNC doesn't line up with the song	The orbits realign when the transport starts, loops or jumps. Start playback from a bar line, and check the host sends tempo and time signature.
The output went silent for a moment	If a sample ever becomes invalid, JVerb outputs silence and restarts its tank (at most once per block) rather than passing a burst of noise. Usually an invalid sample came in from upstream.
My user presets are missing	Check the preset folder for your OS (Chapter 7). Files must end in `.jvpreset`.

Glossary

RT60 The time a reverb takes to fall by 60 dB. JVerb's Decay is the RT60 of the tank's mid frequencies.

AU Astronomical unit. JVerb's playful unit for System Size: at 1 AU the tank's lines run 29 to 202 ms.

Tank The network of delay lines that makes the reverb tail.

Allpass A filter that passes every frequency at the same level but smears it in time. Chains of them build density.

Hadamard mix A way of mixing every line into every other with equal energy, so the tail thickens evenly.

Voice A body's share of the tail, shaped by where that body is.

Slot One of the three places in the signal path the drive or the echo can sit.

Slingshot A close, fast pass of one body by another. It spikes the body's modulation signal.



JVerb

THE THREE-BODY REVERB · V0.7.0

github.com/jeffmhopkins/Jclap